HOUSING MARKET ANALYSIS

Exploring Property Prices, Features and Location Trends

The housing market is influenced by several factors, including property size, location, number of bedrooms and bathrooms, property condition, views, waterfront status, renovation status, basement status and age of the property. Understanding how these factors relate to property prices can provide useful insights into the characteristics associated with property prices and help stakeholders make more informed decisions.

This project analyses a housing dataset obtained from Kaggle to identify patterns and relationships between property prices, property characteristics and location. Microsoft excel was used for initial data inspection and analysis, while Power Query was used for data cleaning and transformation. Power Pivot was subsequently used to create measures, PivotTables and the analytical dashboard.

The analysis focuses on identifying the factors associated with differences in property prices and presenting the findings through tables, charts and key performance indicators.

The analysis was designed to answer the following questions;

1. **What is the overall state of the housing market based on property count, total property value, average price, median price and average price per square foot?**
2. **Which cities have the highest average house prices?**
3. **How does living area relate to average house price?**
4. **How does the number of bedrooms relate to average house price?**
5. **Which cities have the highest average price per square foot?**
6. **How does waterfront status affect average house price?**
7. **How does property view rating relate to average house price?**
8. **How does property condition relate to average house price?**
9. **Do renovated properties have a higher average price than non-renovated properties?**
10. **How does property age relate to average house price?**

Dataset Description

The dataset used for the analysis was obtained from Kaggle and contains information about residential properties. The dataset contains **18 columns** covering property prices, physical characteristics, location and other features. The final Power Pivot analysis reported **4,551 properties** in the analytical model.

Data Cleaning and Transformation Using Excel Power Query

Before conducting the analysis, the dataset was inspected and prepared using Excel Power Query.

The purpose of the cleaning process was to identify unusual values, understand the meaning of zero values and prepare the data for meaningful analysis.

- **Data Quality Checks**
  The columns were reviewed for missing or unusual values. During the inspection, some variables contained zero values. These were investigated rather than automatically removed because a zero could represent a meaning property characteristic.

  For example:
  - A value of 0 in Sqft Basement indicates that the property has no basement.
  - A value of 0 in Year Renovated indicates that the property has not been renovated.
  - Waterfront values of 0 and 1 represent the absence or presence of waterfront access.
  - View and condition values were retained because they represent ratings within the dataset.

  The zero values in the price and price-per-square-foot fields were also identified and reviewed during the data-cleaning process.

Data Transformation

Power Query was used to transform selected variables into more meaningful categories for analysis and visualization.

The following derived columns were created:

Basement Status
The original basement square-footage values were converted into two categories:
- **Has Basement**
- **No Basement**

Waterfront Status
The original waterfront indicator was transformed into:
- **Has Waterfront**
- **No Waterfront**

This made the results easier to interpret than displaying 0 and 1.

Renovation Status
The Year Renovated field was transformed into:
- **Renovated**
- **Not Renovated**

This allowed properties to be compared based on renovation status.

Living Area Band
The living-area values were grouped into ranges to make it easier to examine how property size relates to price.
The categories included ranges such as:
- 0–499 sq ft

- 500–999 sq ft
- 1,000–1,499 sq ft
- 1,500–1,999 sq ft
- 2,000–2,499 sq ft
- 2,500–2,999 sq ft
- 3,000–3,499 sq ft
- 3,500–3,999 sq ft
- 4,000+ sq ft

Property Age

Property age was derived from the year the property was built and grouped into age bands for analysis. The categories included:

- 0–9 years
- 10–19 years
- 20–29 years
- 30–39 years
- 40–49 years
- 50–59 years
- 60+ years

These transformations made the dataset easier to analyse and allowed numerical variables to be presented in a more understandable format.

EXPLORATORY DATA ANALYSIS

Exploratory data analysis was carried out using **Power Pivot measures and PivotTables**. The purpose was to identify patterns, differences and relationships within the housing dataset before presenting the findings visually.

- **Overall Housing Market**

The key summary measures obtained from the Power Pivot model were:

Measure	Result
No. of Properties	4,551
Total Propeerty Value	$3B
Average Price	$558k
Median Price	$465k
Average Price per Sqft	$269

The average property price was considerably higher than the median price. This suggests that the dataset contains a number of high-value properties that increase the overall average.

- **Average House Price by City**

The analysis identified substantial differences in average house prices across cities. The highest average prices included:

- Medina - $2M
- Clyde Hill - $1.6M

- Mercer Island - $1.2M

Other cities among the top locations included Bellevue, Yarrow Point, Seattle and Kirkland.

Finding

Location appears to be an important factor associated with property value, as some cities recorded substantially higher prices than others.

- **Average Price by Living Area**

The analysis showed a general increase in average property price as living area increased.

Some of the results were:

Living Area	Result
0 – 499 sq ft	$206k
500 – 999 sq ft	$296k
1,000 – 1,499 sq ft	$387k
1,500 – 1,900 sq ft	$432k
2,000 – 2,499 sq ft	$549k
2,500 – 2,999 sq ft	$655k
3,000 – 3,499 sq ft	$812k
3,500 – 3,999 sq ft	$953K
4,000+ sq ft	$1.4M

Finding

Larger properties generally recorded higher average prices. Properties with **4000+ square feet** recorded the highest average price among the living-area categories.

- **Average Price by Number of Bedrooms**

The analysis showed that properties with more bedrooms generally recorded average higher prices.

Selected results included:

- 9 bedrooms - $600K
- 8 bedrooms - $1.16M
- 7 bedrooms - $1.05M
- 6 bedrooms - $845K
- 5 bedrooms - $804K
- 4 bedrooms - $643k
- 3 bedrooms - $490K
- 2 bedrooms - $395K
- 1 bedroom - $281K
- 0 bedroom - $1.20M

The 0-bedroom category produced an unusually high average and should be interpreted cautiously because it may be influenced by the characteristics or number of properties within that category.

Finding
There is generally a positive relationship between the bedroom count and average property price, although bedroom count should be considered alongside other characteristics such as property size and location.

- **Average Price per Square Foot by City**

There were notable differences in average price per square foot across cities.
The highest-performing locations included:

- Beaux Arts Village – $500/sqft
- Medina – $492/sqft
- Yarrow Point – $490/sqft
- Clyde Hill – $430/sqft
- Mercer Island – $362/sqft
- Seattle – $326/sqft
- Bellevue – $324/sqft
- Kirkland - $288
- Inglewood - Finn Hill – $280/sqft
- Redmond - $279

Finding
The variation in price per square foot indicates that some locations command a premium even after differences in property size are considered.

- **Average House Price by Waterfront**

A significant difference was observed between properties with and without waterfront access.

Waterfront Status	Average Price
Has Watefront	$1.6M
No Waterfront	$551k

Finding
Properties with waterfront access recorded a substantially higher price than properties without waterfront access. This indicates that waterfront access represents a significant premium feature within the analyzed housing market.

- **Average Price by View Rating**

The analysis showed a clear relationship between view rating and average property price.

View Rating	Average Price
0	$515K
1	$867K
2	$825K
3	$1M
4	$1.3M

Finding

Average property price increased as view rating improved. Properties with the highest view rating recorded significantly higher average prices than properties with no recorded view. This suggests that desirable views may contribute to a property price premium.

- **Average Price by Property Condition**

  The analysis of property condition showed differences in average price across the condition ratings.

Condition	Average Price
1	$307KK
2	$335K
3	$555K
4	$540K
5	$652K

Finding

Properties with higher condition ratings generally recorded higher average prices. Properties rated 5 recorded the highest average price, while properties rated 1 and 2 recorded considerably lower averages.

- **Average Price by Renovation Status**

  The comparison between renovated and non-renovated properties produced the following results

Renovation Status	Average Price
Renovated	$538k
Not Renovated	$571K

Finding

Interestingly, non-renovated properties recorded a slightly higher average price than renovated properties. However, this should not be interpreted as evidence that renovation reduces property value. Other factors such as location, property size, age and condition may influence the result.

- **Average Price by Property Age**

  The analysis of property age produced the following results:

Property Age	Average Price
0 – 9 years	$583K
10 – 19 years	$647K
20 – 29 years	$610K
30 – 39 years	$511K
40 – 49 years	$481K
50 – 59 years	$480K
60+ years	$558K

Finding

The relationship between property age and price was not completely linear. Properties in the **20 – 29 years category recorded the highest average price,** while several older categories lower averages. This suggests that property age alone does not determine property value.

Measures Created In Power Pivot

To support the analysis, calculated measures were created in Power Pivot. The main measures were:

1. Property Count: used to determine the number of properties included in the analysis.
Result: 4,551

2. Total property Value: used to determine the combined value of all properties in the dataset.
Result: $3B

3. Average Price: used to calculate the average selling of properties.
Result: $558K

4. Median Price: used to determine the middle property price and provide a measure less affected by extremely high or low property prices.
Result: $465K

5. Average Price Per Sqft: used to determine the average price paid per square foot
Result: $269

These measures were then used in PivotTables and dashboard visualizations to support the analysis.

Data Visualization and Dashboard

The results of the exploratory analysis were presented through a dashboard created in **Excel using Power Pivot.**

The dashboard was designed to provide a concise overview of the housing market and allow users to compare property prices and across different characteristics.

The dashboard contains five key performance indicators:

- Number of Properties
- Total Property Value
- Average Price
- Median Price

- Average Price per Sqft

It also contains visualizations showing:

- Top 10 Cities by Average House Price
- Average Price by Living Area
- Average Price by Number of Bedrooms
- Top 10 Cities by Average Price per Sqft
- Average House Price by Waterfront Status
- Average House Price by Renovation Status

Interactive slicers were also incorporated for:

- Basement Status
- View
- Condition

These slicers allow users to filter the dashboard and examine how the selected characteristics affect the displayed results.

Key Insights

The analysis produced several important insights:

1. Location is strongly associated with property price.
Cities such as Medina, Clyde Hill and Mercer Island recorded significantly higher average house prices than many other locations.

2. Larger properties generally command higher prices.
Average property price increased considerably as living area increased, with properties above 4,000 square feet recording an average price of approximately **$1.43M**.

3. Bedroom count generally corresponds with higher property prices.
Properties with more bedrooms generally recorded higher average prices, although other factors also influence this relationship.

4. Waterfront properties command a substantial premium.
Waterfront properties recorded an average price of approximately **$1.6M**, compared with approximately **$551K** for properties without waterfront access.

5. Better views are associated with higher property prices.
Average price increased from approximately **$515K** for properties with a view rating of 0 to approximately **$1.3M** for properties with a view rating of 4.

6. Property condition is associated with price.

Properties with higher condition ratings generally recorded higher average prices.

7. Renovation status produced an unexpected result.
Non-renovated properties recorded a slightly higher average price than renovated properties. This indicates that renovation status should not be analysed independently of other property characteristics.

8. Property age does not have a simple linear relationship with price.
The 20 – 29 year property group recorded the highest average price, demonstrating that newer properties are not necessarily always more expensive.

9. Price per square foot varies considerably by location.
Some cities recorded substantially higher prices per square foot, indicating that location can command a premium independent of property size.

Recommendations

Based on the findings, the following recommendations are made:

1. Consider location when evaluating property investments.
Buyers and investors should conduct location-specific analysis because both average house prices and price per square foot vary significantly across cities.

2. Use price per square foot when comparing properties.
Price per square foot provides a useful basis for comparing properties of different sizes and identifying locations where buyers are paying a premium.

3. Treat waterfront properties as a premium market segment.
The substantial difference between waterfront and non-waterfront properties suggests that waterfront access is a valuable feature that should be considered when evaluating property investments.

4. Consider property condition when determining value.
Property condition should be assessed alongside location, size and other characteristics when determining whether a property represents good value.

5. Evaluate renovation based on expected return.
Property owners should not assume that renovation will automatically result in a higher selling price. Renovation decisions should consider the property's location, age, condition and expected market demand.

6. Evaluate multiple property characteristics together.
Buyers, sellers and investors should consider location, living area, bedrooms, views, waterfront access, condition, renovation status and age rather than relying on a single characteristic when assessing property value.

Conclusion

This project analyzed housing market data to identify patterns and relationships between property prices, property characteristics and location.
The analysis revealed considerable differences in property prices across cities and property characteristics. Location, living area, bedroom count, waterfront access, view and property condition were associated with notable differences in average property prices.

Waterfront properties demonstrated a particularly strong price premium, while larger properties and properties with better views generally recorded higher average prices. The analysis also showed that renovation status and property age do not have straightforward relationships with property price and should therefore be considered alongside other factors.

Overall, the analysis demonstrates how data cleaning, transformation, exploratory analysis, data modelling and visualization can be combined to generate meaningful insights from housing market data. The findings can support more informed decision-making by buyers, sellers and investors when evaluating property characteristics and potential value.